

# Status

## Feature Status Ledger

**Revision:** 7 **Last modified:** 2026-06-16T08:32:00Z **Authority:** constitution §11.4.153 (per-feature Status + video-recording confirmation), composes §11.4.44 (revision header), §11.4.45 (integration-status-doc + closed status vocabulary), §11.4.56 (Status\_Summary two-audience companion), §11.4.65 (HTML+PDF export) + §11.4.153 DOCX, §11.4.107 (no frozen-frame video proof), §11.4.6 (no-guessing — every row reconciled against real code). **Scope:** Full feature surface of the ytdlp Podman/Docker orchestration around yt-dlp — every service, client surface, CLI script, landing route, dashboard component, test/challenge suite, and every PLANNED-but-unbuilt feature. **Maintainer:** project conductor.

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## How to read this ledger

- **Implementation** — Implemented (code present + wired), Partial (code present, incomplete/edge-gaps), Planned (not yet built).
- **Validation/Verification** — closed status vocabulary per §11.4.45: PASS / FAIL / SKIP / PENDING\_FORENSICS / OPERATOR-BLOCKED. Because no clean-baseline full-suite run with captured evidence was executed in the session that produced this ledger, verdicts that would require live captured runtime evidence are recorded PENDING\_FORENSICS (per §11.4.6 — an unproven PASS is forbidden; the code is present and test files exist, but a green-with-captured-evidence run is not yet on record here). Planned rows carry SKIP (no feature to validate yet).
- **Video-Confirmation** — per §11.4.153 every user-visible confirmed claim MUST be backed by a recorded real-use capture. The strong-model analysis path is the agent's OWN native multimodal read (Claude Opus 4.8 via the Read tool — see docs/research/vision-path/FINDINGS.md); local CPU models (moondream) are NOT used (too slow + hallucinate). **NINE render/transcript captures are confirmed — dashboard / (download-form + navbar), dashboard /queue, dashboard /history, dashboard /cookies, landing 'Боџа' UI, MeTube UI, postprocess status API, ./status CLI, ./download CLI (together confirming the dashboard's five components + the other surfaces) — AND ONE end-to-end FLOW is confirmed (download→webready transcode, via advancing live-pipeline counter + ffpprobe-validated just-produced artifact).** Every OTHER row remains PENDING — not yet recorded until its window-scoped, ytdlp---prefixed real-use capture (§11.4.154/.155) is recorded + analyzed.

**HONEST FACT (§11.4.6 / §11.4.153):** As of this revision the real PASSes span TEN distinct ytdlp----prefixed captures. NINE are scope=render/transcript: seven image renders analyzed by Claude Opus 4.8 native multimodal — dashboard / (ytdlp---dashboard---20260615T221723Z.png, download-form + navbar), dashboard /queue (ytdlp---dashboard-queue---20260616T083018Z.png, empty state), dashboard /history (ytdlp---dashboard-history---20260616T083023Z.png, content state, 46 items), dashboard /cookies (ytdlp---dashboard-cookies---20260616T083029Z.png, content state, per-platform), landing 'Боба' UI (ytdlp---landing---20260616T074850Z.png), MeTube UI (ytdlp---metube---20260616T075608Z.png), postprocess status API (ytdlp---postprocess---20260616T075648Z.png) — plus two CLI transcripts: ./status (ytdlp---status---20260616T075901Z.txt) and ./download (ytdlp---download-cli---20260616T080936Z.txt). ONE is scope=FLOW: the download→webready transcode (ytdlp---webready-flow---20260616T080633Z.txt), confirmed by the live pipeline's done counter advancing 25→26 across two real status snapshots plus ffprobe-validation of the just-produced 1920×1080 h264 +faststart artifact (the postprocessor is a backend with no UI, so the §11.4.69/.107 sink-side evidence is the validated artifact, not a screen recording). **Still NOT flow-confirmed:** MP3-derivation and cookie-upload FLOWS (the latter operator-gated — needs fresh operator cookies; uploading test cookies would overwrite real operator state, §9.2/§11.4.122; the /cookies render shows YouTube + X cookies expired 2d ago, corroborating the gate). Every other cell honestly reads PENDING – not yet recorded. No row over-claims.

## OPERATOR-BLOCKED rows (top per §11.4.45)

No feature in this project is currently OPERATOR-BLOCKED. (The VPN profile requires operator-supplied .ovpn credentials and vpn-auth.txt, but the VPN *code path* is implemented and testable in compose-config form without live credentials, so it is classified PENDING\_FORENSICS, not OPERATOR-BLOCKED.)

## 1. SERVICES (docker-compose.yml)

| Feature                                                                                           | Component              | Category              | Implementation            | Wiring                                                                      |
|---------------------------------------------------------------------------------------------------|------------------------|-----------------------|---------------------------|-----------------------------------------------------------------------------|
| VPN tunnel (dperson/openvpn-client, NET_ADMIN, /dev/net/tun, port 3130, healthcheck ping 8.8.8.8) | openvpn-ytdlp          | Networking / Service  | Implemented (vpn profile) | Reachable: metube + yt-dl its netns via network_mode service:openvpn-yt-dlp |
| Download queue/history API + minimal UI                                                           | metube / metube-direct | Backend API / Service | Implemented               | Reachable: dashboard nginx to :8081; \${DOWNLOAD_DIR} config volumes        |

| Feature                                                                                                                      | Component                    | Category               | Implementation                    | Wiring                                                                    |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------|-----------------------------------|---------------------------------------------------------------------------|
| (ghcr.io/alexta69/metube:latest, port 8088 direct / netns under VPN)                                                         |                              |                        |                                   |                                                                           |
| Flask 'Боџа' cookie-auth onboarding + aborted-history ledger + proxy endpoints (landing/, ports 8087 vpn / 8086 no-vpn)      | landing-vpn / landing-no-vpn | Web UI + API / Service | Implemented                       | Reachable: published port; delete-download to MeTube aborted.json         |
| Sleeping CLI container for ./download (ghcr.io/jim60105/yt-dlp:pot, while true; do sleep 3600; done, cookie copy at startup) | yt-dlp-cli / yt-dlp-cli-vpn  | CLI backend / Service  | Implemented                       | Reachable: ./download executable; yt-dlp/config, cookie downloads volumes |
| Angular 17 + nginx dashboard (dashboard/, port 9090, /api/* proxied to MeTube + landing)                                     | dashboard                    | Web UI / Service       | Implemented (no-vpn profile only) | Reachable: nginx resolver proxy_pass from entryp                          |
| Image auto-update (containrrr/watchtower:latest, schedule 0 0 */3 * * *, cleanup)                                            | watchtower                   | Infra / Service        | Implemented (docker profile)      | Reachable: docker.sock mount com.centurylinklabs.watchtower               |

## 2. CLIENT SURFACES

| Feature                                                                                          | Component | Category   | Implementation | Wiring                                               | Tests Coverage                                               |
|--------------------------------------------------------------------------------------------------|-----------|------------|----------------|------------------------------------------------------|--------------------------------------------------------------|
| Dashboard web UI (download-form / queue / history / cookies / navbar, error-boundary, not-found) | dashboard | Web client | Implemented    | Routed via app.routes.ts; calls /api/* through nginx | tests/test-dashboard e2e/tests/dashboard component *.spec.ts |
| Landing Flask 'Боџа'                                                                             | landing   |            | Implemented    | INDEX_TEMPLATE in landing/                           |                                                              |

| Feature                                                                          | Component | Category   | Implementation       | Wiring                                                 | Tests Coverage                                          |
|----------------------------------------------------------------------------------|-----------|------------|----------------------|--------------------------------------------------------|---------------------------------------------------------|
| UI (3-step cookie onboarding, drag-drop upload, freshness banner, services grid) |           | Web client |                      | app.py; auto-redirect to dashboard                     | tests/e2e/tests/landing.spec.ts, tests/e2e/dashboard.sh |
| MeTube minimal UI (vendor original interface, dark theme)                        | metube    | Web client | Implemented (vendor) | Served on :8088; linked as "Classic UI" from landing   | tests/test-media-                                       |
| ./download CLI (host script execs yt-dlp in sleeping container)                  | download  | CLI client | Implemented          | ./download → exec into yt-dlp-cli; runtime auto-detect | tests/test-media-challenges/scriptdownload_complete     |

### 3. ROOT CLI SCRIPTS

| Feature                                                | Component | Category   | Implementation | Wiring                       | Tests Coverage                                   |
|--------------------------------------------------------|-----------|------------|----------------|------------------------------|--------------------------------------------------|
| Init environment (create dirs, validate .env, generate | ./init    | CLI script | Implemented    | Runtime auto-detect via lib/ | tests/test-unit.sh, tests/integration.sh, tests/ |

| Feature                                                         | Component      | Category   | Implementation | Wiring                                      | Tests Coverage                                     |
|-----------------------------------------------------------------|----------------|------------|----------------|---------------------------------------------|----------------------------------------------------|
| yt-dlp.conf, write vpn-auth.txt chmod 600)                      |                |            |                | container-runtime.sh                        |                                                    |
| Start services (reads USE_VPN, pulls images, brings up profile) | ./start        | CLI script | Implemented    | compose up via detected runtime             | tests/test-scenarios integration.sh                |
| Start no-vpn (force no-vpn profile)                             | ./start_no_vpn | CLI script | Implemented    | compose --profile no-vpn                    | tests/test-scenarios scripts/no_vpn_profile_direct |
| Stop services (all profiles, clean pods/networks)               | ./stop         | CLI script | Implemented    | compose down + cleanup                      | tests/test-scenarios                               |
| Status summary (service + HTTP health)                          | ./status       | CLI script | Implemented    | queries running containers + HTTP endpoints | tests/test-integrati                               |
| Download helper (yt-dlp via exec)                               | ./download     | CLI script | Implemented    | exec into yt-dlp-cli                        | tests/test-media-ser scripts/download_com          |
| Check VPN (query ipinfo.io inside VPN container)                | ./check-vpn    | CLI script | Implemented    | exec into openvpn-yt-dlp                    | tests/test-vpn-smoke                               |
| Restart services                                                | ./restart      | CLI script | Implemented    | stop + start sequence                       | tests/test-scenarios                               |

| Feature                                                        | Component            | Category   | Implementation | Wiring                                   | Tests Coverage                                    |
|----------------------------------------------------------------|----------------------|------------|----------------|------------------------------------------|---------------------------------------------------|
| Cleanup containers/pods/networks                               | ./cleanup            | CLI script | Implemented    | compose down + prune                     | tests/test-scenarios/scripts/container_restart_re |
| Release preparation (commit + push)                            | ./prepare-release.sh | CLI script | Implemented    | git operations                           | bash -n syntax gate (make                         |
| Auto-update setup (cron for Podman; Watchtower alt for Docker) | ./setup-auto-update  | CLI script | Implemented    | writes cron entry / documents Watchtower | bash -n syntax gate                               |
| Update images (pull latest before start)                       | ./update-images      | CLI script | Implemented    | compose pull via detected runtime        | bash -n syntax gate                               |

## 4. LANDING ROUTES (landing/app.py)

| Feature                                                                                                  | Component | Category         | Implementation | Wiring                                                        |
|----------------------------------------------------------------------------------------------------------|-----------|------------------|----------------|---------------------------------------------------------------|
| / — BoBa onboarding page (3-step cookie flow, session id, dynamic service URLs)                          | landing   | HTTP route       | Implemented    | render_template_string(INDEX_TE                               |
| /app — proxy GET to MeTube backend                                                                       | landing   | HTTP proxy route | Implemented    | requests.get(METUBE_URL) streamec                             |
| /api/upload-cookies (POST) — validate Netscape file (UTF-8 + recognised-domain gate) → forward to MeTube | landing   | HTTP API route   | Implemented    | _validate_cookie_file + POST to \${METUBE_URL}/upload-cookies |
|                                                                                                          | landing   |                  | Implemented    | os.remove                                                     |

| Feature                                                                                     | Component | Category       | Implementation | Wiring                                           |
|---------------------------------------------------------------------------------------------|-----------|----------------|----------------|--------------------------------------------------|
| /api/delete-cookies (POST) — remove /config/cookies.txt                                     |           | HTTP API route |                |                                                  |
| /api/cookie-status (GET) — has_cookies + age + per-platform breakdown + MeTube reachability | landing   | HTTP API route | Implemented    | _summarize_cookies_by_platform + MeTube fallback |
| /api/aborted-history (GET) — return persisted aborted list                                  | landing   | HTTP API route | Implemented    | _read_aborted_history (lock-free re              |
| /api/aborted-history (POST) — append entry, flock-guarded, 60s idempotency                  | landing   | HTTP API route | Implemented    | _aborted_history_lock + _write_aborted_history   |
| /api/aborted-history (DELETE) — remove specific urls or *                                   | landing   | HTTP API route | Implemented    | locked read-modify-write                         |
| /api/profile-status (GET) — active compose profile + vpn_active + MeTube reachability       | landing   | HTTP API route | Implemented    | reads ACTIVE_PROFILE env                         |
|                                                                                             | landing   |                | Implemented    | requests.get('\${METUBE_URL}/hist                |

| Feature                                                                                                                         | Component | Category                | Implementation | Wiring                                                             |
|---------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------|----------------|--------------------------------------------------------------------|
| /health<br>(GET) —<br>service +<br>MeTube<br>reachability +<br>timestamp                                                        |           | HTTP<br>API route       |                |                                                                    |
| /logo.png<br>(GET) —<br>serve<br>branding<br>asset                                                                              | landing   | HTTP<br>static<br>route | Implemented    | send_from_directory                                                |
| /favicon.ico<br>(GET) — 204<br>stub                                                                                             | landing   | HTTP<br>static<br>route | Implemented    | returns "", 204                                                    |
| /api/<br>delete-<br>download<br>(POST) —<br>remove from<br>history +<br>optional file<br>delete (path-<br>traversal<br>guarded) | landing   | HTTP<br>API route       | Implemented    | POST \${METUBE_URL}/delete + os.w<br>os.remove within DOWNLOAD_DIR |

## 5. DASHBOARD COMPONENTS (dashboard/src/app)

| Feature                                                                                                                                                  | Component     | Category             | Implementation | Wiring                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------|----------------|------------------------------------|
| Download form (url,<br>quality select, format<br>select, folder, add via /<br>api/add, inline tracker +<br>error state, re-enable after<br>failure)      | download-form | Angular<br>component | Implemented    | MeTubeService.a                    |
| Queue (STATE_META:<br>pending / preparing /<br>downloading /<br>postprocessing / finished /<br>error / aborted; cancel-all;<br>record-and-close aborted) | queue         | Angular<br>component | Implemented    | getHistoryPolli<br>recordAbortedIt |

| Feature                                                                                                             | Component                 | Category          | Implementation | Wiring                                                  |
|---------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------|----------------|---------------------------------------------------------|
| History (+ aborted-history merge via abortedToDownloadInfo, retry, delete-with-file, batch + delete-all, refresh)   | history                   | Angular component | Implemented    | getHistory + getAbortedHistory + deleteDownloadWithFile |
| Cookies (status badge fresh/expiring/expired, age, MeTube online/offline, per-platform breakdown, upload, delete)   | cookies                   | Angular component | Implemented    | getCookieStatus + uploadCookies + deleteCookies         |
| Navbar (routerLinks Download/Queue/History/Cookies; VPN / No-VPN / unknown profile badge; MeTube-unreachable badge) | navbar                    | Angular component | Implemented    | getProfileStatus + getVPNStatus                         |
| Error boundary (renders loading / error+retry / empty / content states)                                             | error-boundary            | Angular component | Implemented    | wrapped around list components                          |
| Not-found (404 catch-all route)                                                                                     | not-found                 | Angular component | Implemented    | ** route in app.routes                                  |
| Error interceptor (RxJS error surfacing, clears loading)                                                            | error-interceptor.service | Angular service   | Implemented    | app.config.ts provide                                   |
| MeTube service (typed API: add/start/history/delete/clear/retry/cookies/profile/aborted/version/poll; no any)       | metube.service            | Angular service   | Implemented    | HttpClient to /api                                      |

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## 6. TESTS / CHALLENGES (real test files reconciled)

| Feature                                                      | Component                                                   | Category   | Implementation | Wiring                      |
|--------------------------------------------------------------|-------------------------------------------------------------|------------|----------------|-----------------------------|
| Unit tests (pure-shell logic)                                | tests/test-unit.sh                                          | Test suite | Implemented    | run-tests.sh -p             |
| Integration tests (containers up)                            | tests/test-integration.sh                                   | Test suite | Implemented    | run-tests.sh -p integration |
| Real-HTTP integration (no mocks, hits :9090/:8086/:8088)     | tests/test-integration-realhttp.sh                          | Test suite | Implemented    | services must be up         |
| Scenario tests (lifecycle phases)                            | tests/test-scenarios.sh                                     | Test suite | Implemented    | run-tests.sh -p scenario    |
| Error-phase tests                                            | tests/test-errors.sh                                        | Test suite | Implemented    | run-tests.sh -p             |
| Dashboard + landing integration                              | tests/test-dashboard.sh                                     | Test suite | Implemented    | services up                 |
| Dashboard operations (post-fix full automation)              | tests/test-dashboard-operations.sh                          | Test suite | Implemented    | services up                 |
| Aborted-history (incl. concurrent-posts no-500)              | tests/test-aborted-history.sh                               | Test suite | Implemented    | landing up                  |
| Cookie validator (_validate_cookie_file)                     | tests/test-cookie-validator.sh                              | Test suite | Implemented    | landing app.py              |
| Media-services compatibility (anti-bluff documented-failure) | tests/test-media-services.sh                                | Test suite | Implemented    | yt-dlp-cli / metube u       |
| Bulk operations                                              | tests/test-bulk-operations.sh                               | Test suite | Implemented    | services up                 |
| Add-all-platforms / add-download                             | tests/test-add-all-platforms.sh, tests/test-add-download.sh | Test suite | Implemented    | metube up                   |
| VPN compose + smoke                                          | tests/test-vpn-compose.sh,                                  | Test suite | Implemented    | vpn profile                 |

| Feature                                                                                                                                                                                                                                    | Component                                                                           | Category                        | Implementation          | Wiring                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------|-------------------------|---------------------------------------------------|
|                                                                                                                                                                                                                                            | tests/test-vpn-smoke.sh                                                             |                                 |                         |                                                   |
| Chaos / resilience                                                                                                                                                                                                                         | tests/test-chaos.sh                                                                 | Test suite                      | Implemented             | services up                                       |
| Constitution inheritance + meta-test                                                                                                                                                                                                       | tests/test-constitution-inheritance.sh, tests/meta-test-constitution-inheritance.sh | Test suite                      | Implemented             | repo tree                                         |
| E2E (Playwright): dashboard, landing, cross-service, clear-history, delete-retry-cancel, cleanup-and-start                                                                                                                                 | tests/e2e/                                                                          | E2E suite                       | Implemented             | run-e2e.sh + playwright.conf                      |
| Benchmark suite                                                                                                                                                                                                                            | tests/benchmark/run-benchmarks.sh                                                   | Benchmark suite                 | Implemented             | services up                                       |
| Cookie auth + youtube-download + validate-fix                                                                                                                                                                                              | tests/cookies/                                                                      | Test suite                      | Implemented             | metube + cookies                                  |
| MeTube anti-bluff challenge bank (download-completes, queue-realtime, form-reenables)                                                                                                                                                      | tests/challenges/                                                                   | Challenge bank                  | Implemented             | run-metube-challenges.sh + metube-challenges.json |
| Root HelixQA challenge scripts (api-contract, container-restart-resilience, download-completes, form-reenables, host/user OOM+suspend, landing-cookie-upload, memory-limits, no-vpn-direct, queue-lifecycle, queue-polling, retry-visible) | challenges/scripts/                                                                 | Challenge bank                  | Implemented             | run_all_challeng                                  |
| Challenges submodule (HelixQA challenge framework, p1-f06..f18 CLI-agent feature dirs)                                                                                                                                                     | Challenges/                                                                         | Challenge framework (submodule) | Implemented (submodule) | Challenges/ catalo                                |

# 7. MEDIA POST-PROCESSOR

## (media\_postprocessor — shipped ytdlp-1.4.0)

Reconciled 2026-06-16 (Rev 4): these rows were marked Planned in Rev 3 but the dual-version pipeline shipped in tag ytdlp-1.4.0. Verified as FACT: real Python package media\_postprocessor/ (config/jobs\_db/media\_probe/transcoder/watcher/worker/service), compose service media\_postprocessor (container media-postprocessor, MP\_PORT=8089, MP\_DB\_PATH=/downloads/.media\_postprocessor/jobs.db), nginx proxy dashboard/nginx.conf.template:116 /api/postprocess/ → media-postprocessor:8089, live endpoint returning real JSON, 6+ real webready-\*.mp4 on disk.

| Feature                                                                                                                                  | Component           | Category         | Implementation            | Wiring                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Media post-processor sidecar (SQLite-WAL queue, watcher + worker, crash-safe resume, atomic output, ffmpeg-validate, /postprocess/* API) | media_postprocessor | Service          | Implemented (ytdlp-1.4.0) | Reachable: compose service media_postprocessor (cont media-postprocessor, MP_PORT=8089); dashboard n proxies /api/postprocess/ media-postprocessor:8089 (nginx.conf.template:116 |
| Web-ready video transcode (webready-<base>.mp4, H.264/AAC, +faststart)                                                                   | media_postprocessor | Pipeline feature | Implemented               | transcoder.transcode_vi → webready-<base>.mp4, ffmpeg-validate (_validate_webready assert h264 + faststart); watcher(+m age) + worker                                            |
| MP3 audio derivation (<base>.mp3, 320 kbps CBR)                                                                                          | media_postprocessor | Pipeline feature | Implemented               | transcoder.derive_mp3() <base>.mp3; worker invokes audio jobs                                                                                                                    |
| Post-process status API (/                                                                                                               | media_postprocessor |                  | Implemented               |                                                                                                                                                                                  |

| Feature                                                     | Component | Category       | Implementation | Wiring                                                                          |
|-------------------------------------------------------------|-----------|----------------|----------------|---------------------------------------------------------------------------------|
| postprocess/<br>status —<br>healthy + per-<br>state counts) |           | API<br>feature |                | media_postprocessor.serv<br>postprocess/status; nginx<br>api/postprocess/ proxy |

## 8. PLANNED — not yet built (Implementation = Planned)

Confirmed ABSENT from the codebase, re-verified 2026-06-16 (Rev 4): no matching Angular component (find dashboard/src/app -iname '\*postprocess\*' -o -iname '\*pipeline\*' → none) and no user-facing download-resume feature — §11.4.6.

| Feature                                                                              | Component                       | Category                         | Implementation | Wiring                                                                                                                                      | Tests Coverage | Val Ver |
|--------------------------------------------------------------------------------------|---------------------------------|----------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------|
| Dashboard<br>pipeline-<br>status UI<br>(surface<br>transcode/<br>derive<br>progress) | dashboard                       | Web UI<br>feature<br>(planned)   | Planned        | Not wired —<br>no component<br>exists                                                                                                       | None yet       | SK      |
| Resume<br>mechanism<br>(resume<br>interrupted/<br>aborted<br><i>downloads</i> )      | metube /<br>media_postprocessor | Pipeline<br>feature<br>(planned) | Planned        | Not wired (the<br>postprocessor's<br>crash-safe <i>job</i><br>resume is<br>implemented;<br>user-facing<br><i>download</i><br>resume is not) | None yet       | SK      |

## Reconciliation note (§11.4.153 / §11.4.118)

Every row above maps to real code (a named compose service, a Flask route in landing/app.py, an Angular component/service file under dashboard/src/app/, a root CLI script, a media\_postprocessor/ module, or a test/challenge file) — except the §8 PLANNED rows, which are explicitly marked Planned/SKIP because the codebase contains no matching implementation (re-verified 2026-06-16). No code-present feature is omitted; no row lacks corresponding code. Per §11.4.6, no verdict claims a PASS without captured evidence. Per §11.4.153, exactly FIVE

Video-Confirmation cells claim a real PASS — dashboard UI, landing 'Боџа' UI, MeTube UI, postprocess status API, and the ./status CLI — each backed by a ytdlp---prefixed captured artifact analyzed by Claude Opus 4.8 native multimodal, **all scope=render/transcript (none flow-confirmed)**; every other cell reads PENDING – not yet recorded.